



CONVERSATIONAL GRAPH EXTRACTION FROM UNSTRUCTURED DATA

PROBLEM STATEMENT

- **Challenge:** Extracting Knowledge Graphs from unstructured data (chats, emails, web, ...).
- **Reality:** Users rarely know the right ontology before exploring the data.
- **Problems of available methods:**
 - Pre-defined schemas, risk making incorrect commitments early on and missing valuable information
 - With unconstrained extraction (OpenIE) we end up with noisy, unqueryable graphs full of duplicated entities

The main idea of this project is: schema design should be an *active, conversational negotiation* over data samples.

PROBLEM SCOPE & RESEARCH QUESTIONS

Scope: Unstructured datasets where multiple valid ontologies exist (e.g., Scientific Papers).

1. **Efficacy:** Does conversational schema discovery produce more useful KGs than zero-shot LLM extraction?
2. **Evaluation:** How do we measure schema quality when there is no objective "ground-truth" ontology?
3. **Convergence:** How can we computationally detect when the schema has stabilized and stop asking the user?

STATE OF THE ART

- **LLMs4SchemaDiscovery (2025)**: A HITL workflow for scientific schema mining via LLMs. Demonstrates the efficacy of expert feedback coupled with LLMs.
- **CLARE: Context-Aware Interactive KG Construction (2025)**: Explores interactive KG construction from transcripts, proving that conversational context improves extraction.
- **Microsoft GraphRAG (2024)**: Validated massive schema-less extraction but lacks a continuous interactive loop for user-guided ontology synthesis.
- **Prompt-guided LLM Agents for Ontology Learning (2025)**: Proposes a loop of "schema generation / fix-instances" to achieve consistency via LLM agents.

INITIAL THOUGHTS ON EXPERIMENTATION

Baseline

Standard zero-shot
LLM extraction

Target Datasets

- SciERC (Scientific papers)
- HackerNews articles

Key Metrics

- Evaluating the generated graph on a specific task such GraphRAG
- Schema stability (cross-validation)

INITIAL SYSTEM ARCHITECTURE

- **LLM Engine:** LiteLLM + Pydantic.
- **Backend:** FastAPI
- **Database:** Neo4j
- **Frontend:** Streamlit or React + Vite

THANK YOU